

BEC601 _ Embedded System Design

Module 1

1. What is embedded system? List the applications of embedded system.
2. With a neat diagram explain the components of typical Embedded Systems in detail.
3. Give the difference between Microcontroller and Microprocessor.
4. Explain the following: i) Opto coupler. ii) Push button switch.
5. Explain the classifications of Embedded System with examples.
6. Give the difference between Van-Neumann and Harvard Architecture.
7. What are the different types of memories used in embedded system design? Explain the role of each?
8. Explain the following with neat diagrams: i) I²C bus ii) SPI bus iii) IrDa iv) Bluetooth
9. Differentiate between (i) General Computing Systems and Embedded Systems and (ii) Little Endian and Big Endian memory access types.

Module 2

1. Explain the characteristics of an Embedded System
2. What are operational quality attribute? Explain the important operational quality attributes to be considered in any embedded system design.
3. Explain the components and working of Application Specific Embedded system- Washing Machine.
4. Explain Hardware and software co-design in embedded system
5. With the help of FSM model, explain the system design and operation of seatbelt warning system.
6. Explain DFG and CDFG models with an example.
7. What are Non operational quality attribute? Explain the non operational quality attributes to be considered in any embedded system design.